12.3. STAR IGNITION

The condensation of matter releases gravitational energy which is transformed into kinetic energy of the particles in the gas cloud, increasing temperature and pressure. As the density increases, radiation transport through the cloud becomes more difficult, leading to more rapid heating of the gas.

At a critical density, the *Chandrasekhar limit*, the hydrogen degenerates into H^+ ions and non-localized ("free") electrons (this is the *plasma state*). The ionization energy of H (13.6 eV) corresponds to an average kinetic particle energy of 160 000 K, eqn. (3.31), but because of the particle energy distribution (Fig. 3.4), degeneration begins already at lower temperature. Because the elimination of the electron shell leads to a pressure reduction, the gravitational forces cause an increased rate of condensation, with a further rise in temperature until eventually all the gas is ionized.

Because the temperature decreases from the center towards the shell, the light emitted from the center may be absorbed in outer layers. Thus the protostar may be invisible, but as it gradually heats up it begins to emit light. There are many such faint objects in our galaxy.

Depending on the mass of the contracting cloud, the star evolution follows different paths.

(i) For a cloud with $m < 0.08\ \mathbf{M}_\odot$, temperature and pressure never reach high enough values for hydrogen ignition and such "stars" will contract to planet-like *brown dwarfs.*

(ii) For masses $>0.08\ \mathbf{M}_\odot$ fusion reactions begin when the core reaches $\geq 4 \times 10^6$ K. In Figure 12.4 we have indicated the birth of a star like our sun ($1\ \mathbf{M}_\odot$) which appears initially as a red object in the sky, indicating a low surface temperature, but the luminosity slowly increases over a condensation time of $10^5 - 10^6$ y.

The Coulomb barrier for interaction of protons with each other is 1.1 MeV, eqn. (10.15). In the sun, density and temperature increases strongly from the surface towards the core. At a core temperature of 1.4×10^7 K, the average kinetic energy of a proton is only 1.8 keV (the most probable energy is 1.2 keV, § 3.6.2, while the fraction of protons with energies $\geq$1.1 MeV is only $\sim 10^{-398}$, eqn. (3.35). Formation of a ^{2}He nucleus from two protons is endothermic and will not occur by tunnelling due to the lack of a driving force. On the other hand the reaction forming ^{2}H is exothermic. Hence, quantum mechanical tunnelling can allow two protons to interact to form a ^{2}H nucleus according to (12.13) during the brief encounter of a scattering event even at energies $\ll$1.1 MeV. The reaction cross section is extremely small due to the involvement of the weak force and a change in spin for one of the nucleons ($\sim 10^{-47}$ barn at 1 MeV collision energy). Once deuterium has been formed it may fuse into helium. Thus the star has begun its "childhood" by burning hydrogen into helium: it has entered the Zero Age Main Sequence of stars. Stars with smaller masses enter lower on the Main Sequence, and evolve very slowly.

(iii) Stars with $0.08\ \mathbf{M}_{\odot} < m < 0.26\ \mathbf{M}_{\odot}$ are completely convective, leading to hydrogen burning to helium, until all hydrogen is exhausted, after which the star contracts to a *white dwarf.*

Other stars, with $m > 0.26\ \mathbf{M}_{\odot}$, evolve differently depending on their masses, as shown in Figure 12.4 and described in the following sections.

12.4. FUSION PROCESSES IN STARS

The elemental composition of our sun is about 73% hydrogen, 25% helium, and 2% carbon, nitrogen, oxygen, and other elements distributed as shown in Figure 12.5. In all, approximately 70 elements have been detected in the solar spectrum and there are reasons to believe that all the elements to uranium are present in our sun. Let us now consider the reactions for the formation of all these elements and the energy producing nuclear processes in our sun and other stars.

The present understanding of processes in the interior of stars is the result of combined efforts from many scientific disciplines such as hydrodynamics, plasma physics, nuclear physics, nuclear chemistry and not least astrophysics. To understand what is going on in the inaccessible interior of a star we must make a model of the star which explains the known data: mass, diameter, luminosity, surface temperature and surface composition. The development of such a model normally starts with an assumption of how elemental composition varies with distance from the center. By solving the differential equations for pressure, mass, temperature, luminosity and nuclear reactions from the surface (where these parameters are known) to the star's center and adjusting the elemental composition model until zero mass and zero luminosity is obtained at the center one arrives at a model for the star's interior. The model developed then allows us to extrapolate the star's evolution backwards and forwards in time with some confidence. Figure 12.7 shows results from such modelling of the sun.

12.4.1. Hydrogen burning to helium

Helium can be formed from hydrogen in several ways, the least likely one is (12.5), which would require that 4 protons come together simultaneously. Similarily to normal chemical rate reactions a reaction of this type involving a multi body reaction it is likely that the reaction is a multi-step process which one rate determining step. The possible multi-step processes are (not showing γ's)

$$^{1}\mathrm{H} + {}^{1}\mathrm{H} \rightarrow {}^{2}\mathrm{H} + \left|\mathrm{e}^{-} + \mathrm{e}^{+}\right| + \nu_{\mathrm{e}} \quad Q = 0.42 + 1.02 = 1.44\ \mathrm{MeV} \tag{12.13}$$

$$^{1}\mathrm{H} + {}^{2}\mathrm{H} \rightarrow {}^{3}\mathrm{He} \qquad Q = 5.49\ \mathrm{MeV} \tag{12.15}$$

$$^{1}\mathrm{H} + {}^{3}\mathrm{H} \rightarrow {}^{4}\mathrm{He} \qquad Q = 19.81\ \mathrm{MeV} \tag{12.16}$$